Ball Bearings Catalogue

Kaveri Motion Components · Rev 2025

About this catalogue

This extract lists the most frequently ordered single row deep groove ball bearings stocked at KMC Chennai.

Bearing selection overview

Select a bearing by shaft diameter, housing bore, load and speed. Sealed bearings suit contaminated environments; shielded bearings suit higher speeds.

Tolerances

Bearings are manufactured to ISO normal tolerance class P0 unless stated otherwise.

Designations and suffixes

Suffix -2RS: rubber contact seals on both sides, greased for life. Suffix -ZZ: metal shields on both sides. No suffix: open bearing. Suffix C3: internal radial clearance greater than normal, used for high operating temperature or when both rings have interference fits. Sealed and shielded bearings are not relubricated.

Dimensions and load ratings

Single row deep groove ball bearings.

6204	bore d 20 mm, outside diameter D 47 mm, width B 14 mm, dynamic load rating C 13.5 kN, static load rating C0 6.55 kN, limiting speed 11,000 rpm for -2RS and 17,000 rpm for -ZZ
6205	bore d 25 mm, outside diameter D 52 mm, width B 15 mm, dynamic load rating C 14.8 kN, static load rating C0 7.8 kN, limiting speed 9,500 rpm for -2RS and 15,000 rpm for -ZZ
6206	bore d 30 mm, outside diameter D 62 mm, width B 16 mm, dynamic load rating C 20.3 kN, static load rating C0 11.2 kN, limiting speed 8,000 rpm for -2RS and 13,000 rpm for -ZZ
6305	bore d 25 mm, outside diameter D 62 mm, width B 17 mm, dynamic load rating C 23.4 kN, static load rating C0 11.6 kN, limiting speed 8,500 rpm for -2RS and 13,000 rpm for -ZZ

Bearing life

Basic rating life L10 = (C/P)^3 million revolutions, where C is the dynamic load rating and P the equivalent dynamic bearing load. A bearing with a higher load rating but different boundary dimensions is not a substitute unless the shaft and housing are changed.